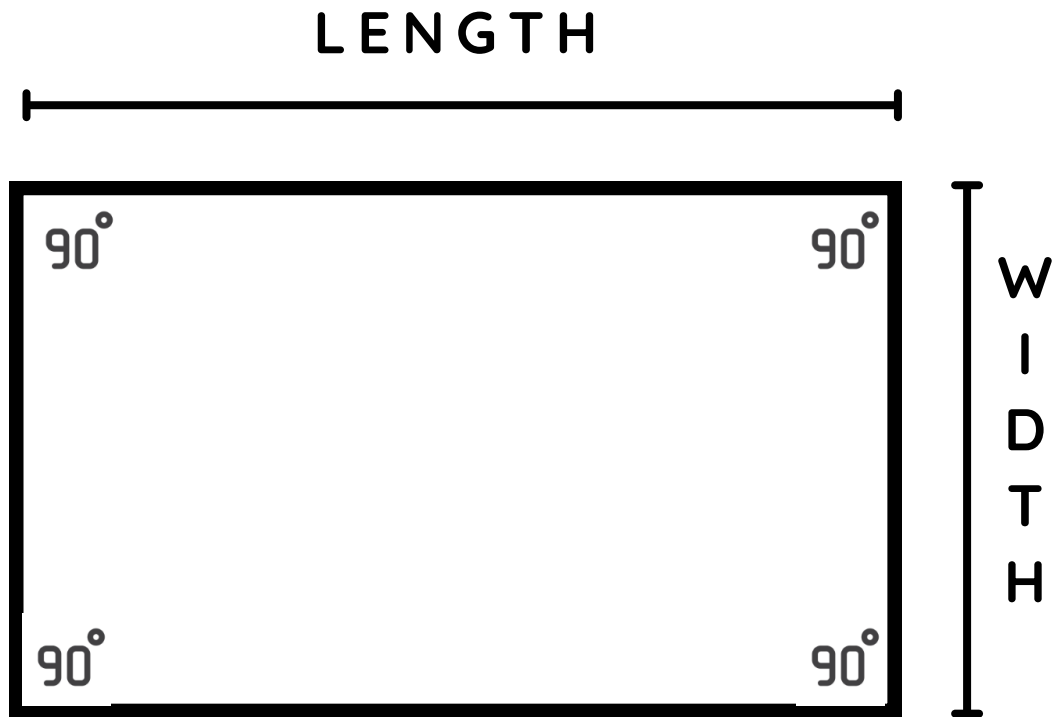


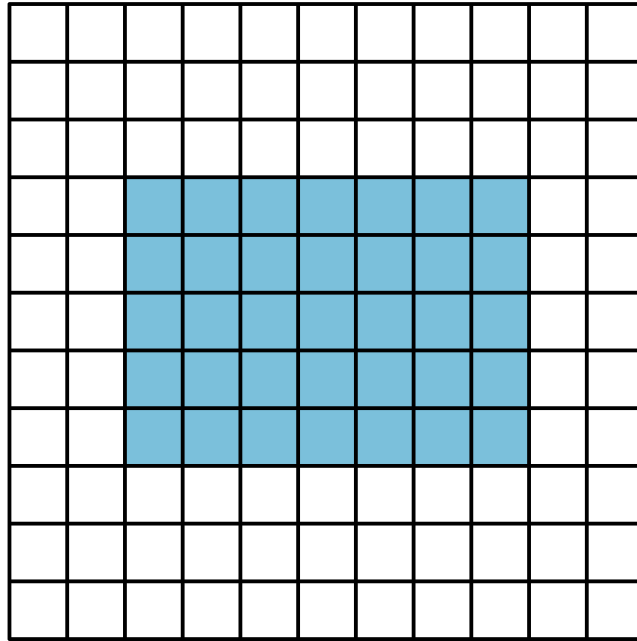
Area and Perimeter of Rectangles

Rectangle



- Has 4 sides.
- Has 4 right angles (each 90 degrees).
- Opposite sides are equal and parallel.
- The longer sides are called the length, and the shorter sides are called the width.
- It is a type of quadrilateral and also a parallelogram.
- The diagonals of a rectangle are equal in length and bisect each other.

Area on a Grid



A grid has vertical and horizontal lines that form a square. Each square measures 1 sq. unit.

Let's count how many squares the blue rectangle covers.

There are 35 squares.

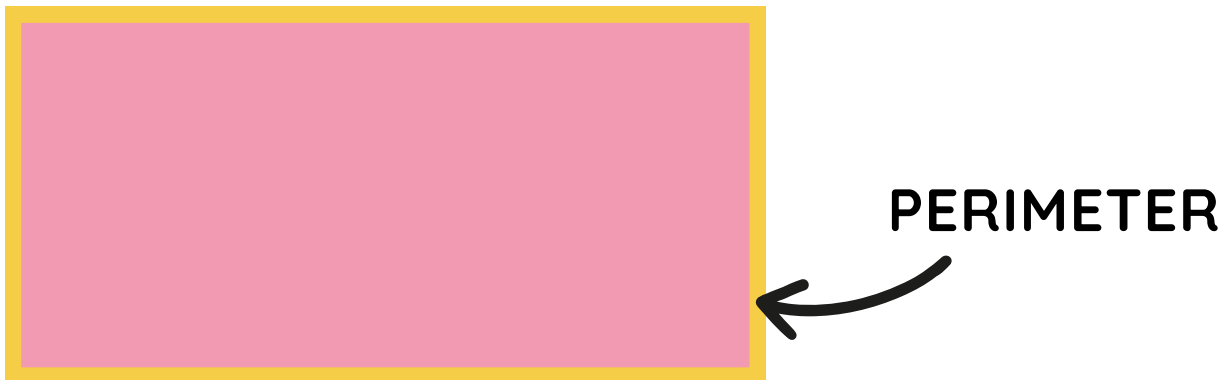
This means that the rectangle has an area of 35 square units.

Area & Perimeter



Area

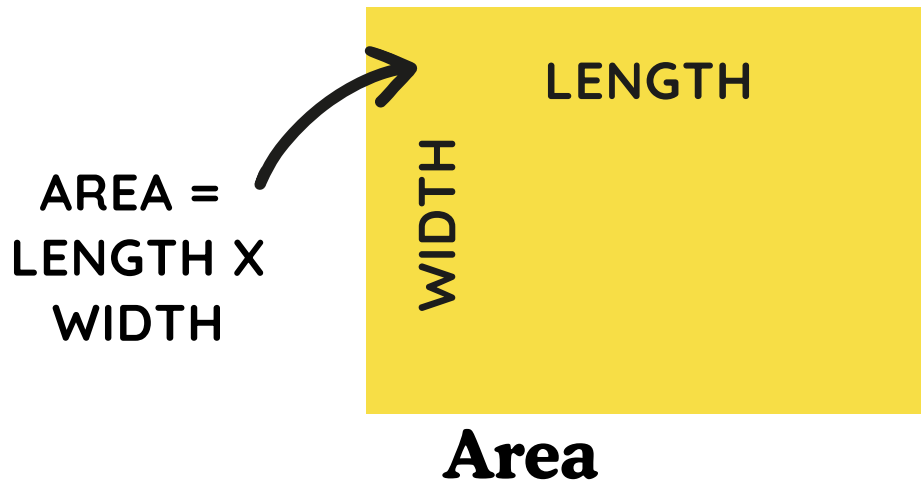
The amount of space inside a shape.



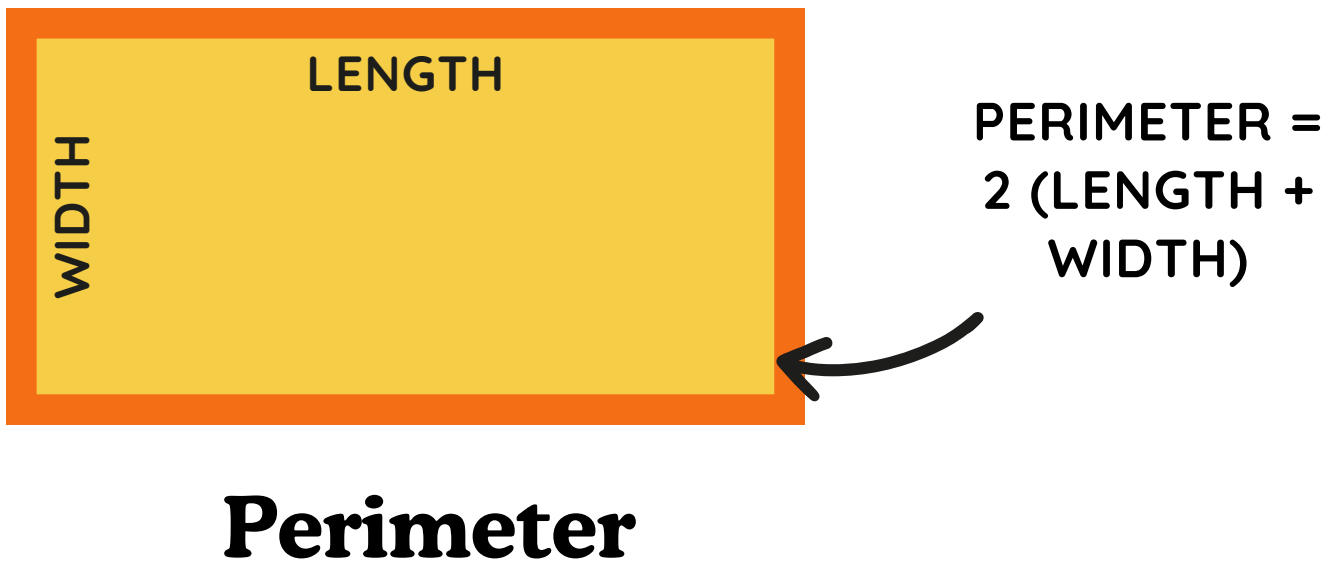
Perimeter

The total distance around the outside of a shape.

Area & Perimeter

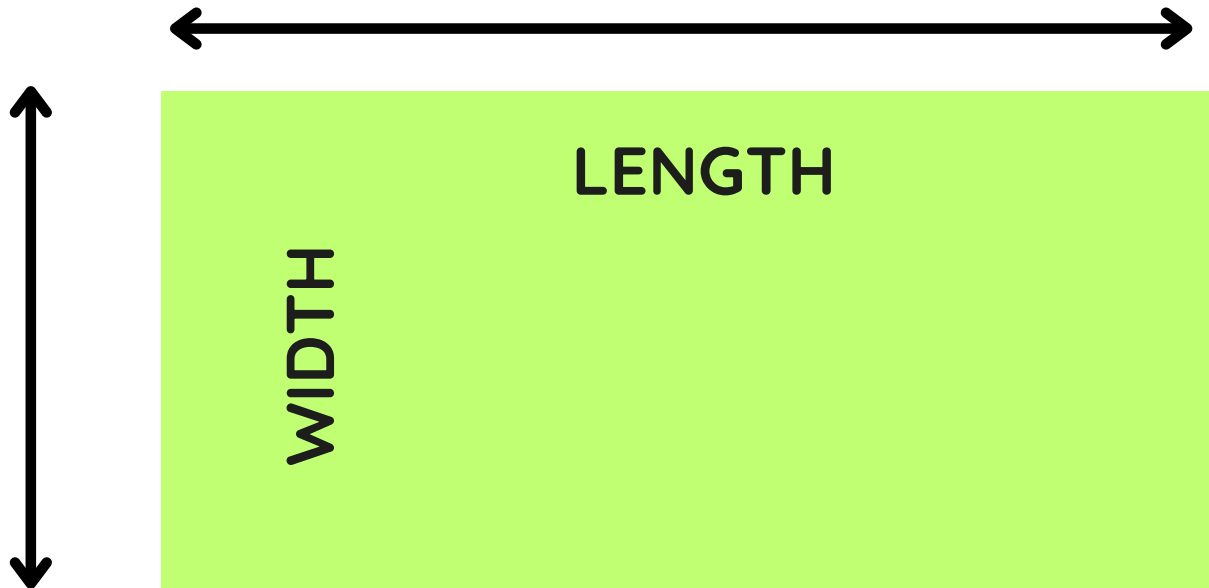


The area is the product of length and width. Area is expressed in square units.



The perimeter is twice the sum of the length and width. Perimeter is expressed in units.

Formulas



Area of Rectangle

$$A = L \times W$$

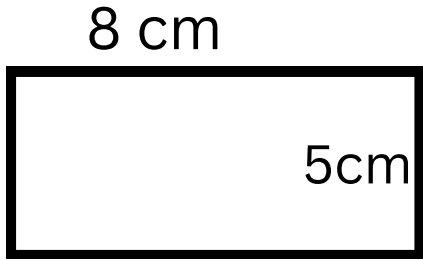
Perimeter of Rectangle

$$P = 2 (L + W)$$

Worksheets

Find Area & Perimeter

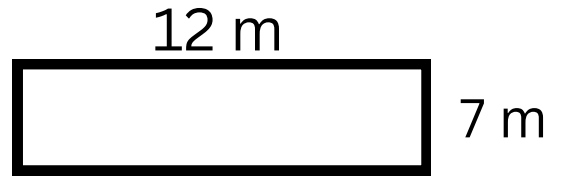
1.



Area = _____ cm^2

Perimeter = _____ cm

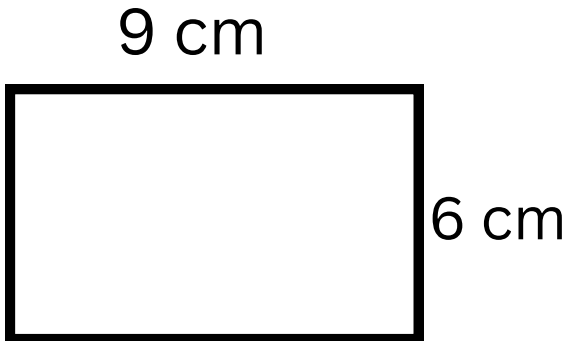
2.



Area = _____ m^2

Perimeter = _____ m

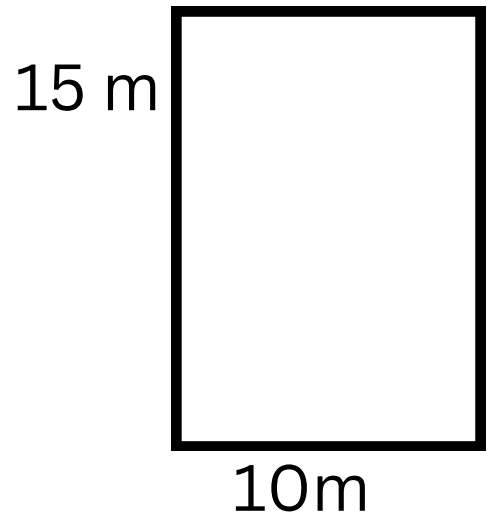
3.



Area = _____ cm^2

Perimeter = _____ cm

4.



Area = _____ m^2

Perimeter = _____ m

Find Area & Perimeter

(continued)

5.

14 cm

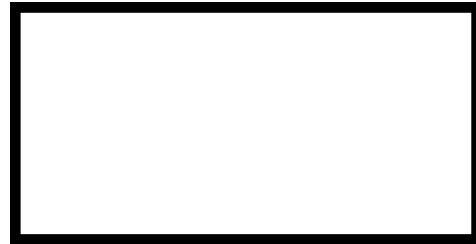


Area = _____ cm^2

Perimeter = _____ cm

6.

11 m



Area = _____ m^2

Perimeter = _____ m

7.

21 cm



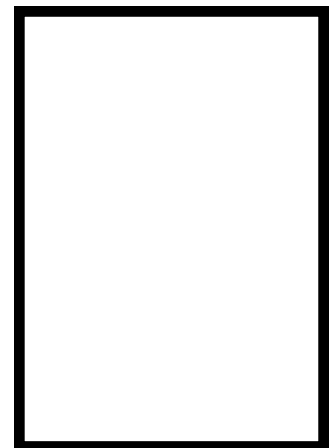
15 cm

Area = _____ cm^2

Perimeter = _____ cm

8.

25 m



13m

Area = _____ m^2

Perimeter = _____ m

Worksheet: Find the Missing Length or Width

1. The area of a rectangle is 42 cm^2 and the length is 7 cm .

What is the width?

Width = _____ cm

2. The perimeter of a rectangle is 30 cm and the width is 5 cm .

What is the length?

Length = _____ cm

3. A rectangle has an area of 72 m^2 and a width of 8 m .

What is the length?

Length = _____ m

4. The perimeter of a rectangle is 44 cm and the length is 12 cm .

Find the width.

Width = _____ cm

Worksheet: Find the Missing Length or Width (continued)

5. The area of a rectangle is 56 m^2 and one side is 7 m.

What is the other side?

Other side = _____ m

6. A rectangle has a perimeter of 36 cm and a width of 6 cm.

What is the length?

Length = _____ cm

7. The area of a rectangle is 90 cm^2 and the length is 10 cm.

Find the width.

Width = _____ cm

8. A rectangle has a perimeter of 50 m and the length is 15 m.

What is the width?

Width = _____ m

Worksheet: Word Problems

1. A garden is in the shape of a rectangle. Its length is 12 m and width is 5 m.

What is the total area of the garden? What is the length of the fence required to go around it?

2. A rectangular table top has a length of 10 cm and width of 6 cm.

What is its area and perimeter?

3. A rectangular carpet is 15 feet long and 8 feet wide.

Find the area of the carpet and the amount of border needed to go around it.

Worksheet: Word Problems

(continued)

4. A swimming pool is in the shape of a rectangle. Its length is 25 m and width is 10 m. What is the area of the pool? How much tile is needed to cover the border (perimeter)?

5. A rectangular classroom is 20 m long and 10 m wide.

What is the area of the floor? What is the perimeter of the classroom?

6. A whiteboard has a length of 9 feet and a width of 4 feet.

Find the area and the perimeter of the whiteboard.

Worksheet: Word Problems

(continued)

7. A rectangular window is 7 ft tall and 3 ft wide. What is the area of glass needed? How much wood is required to frame it?

8. A rectangular park is 30 m long and 20 m wide.

What is the area of the park? What is the perimeter of the walking path around it?

Find Area & Perimeter

(Answers)

1. $8 \text{ cm} \times 5 \text{ cm}$

- $\text{Area} = 8 \times 5 = 40 \text{ cm}^2$
- $\text{Perimeter} = 2 \times (8 + 5) = 2 \times 13 = 26 \text{ cm}$

2. $12 \text{ m} \times 7 \text{ m}$

- $\text{Area} = 12 \times 7 = 84 \text{ m}^2$
- $\text{Perimeter} = 2 \times (12 + 7) = 2 \times 19 = 38 \text{ m}$

3. $9 \text{ cm} \times 6 \text{ cm}$

- $\text{Area} = 9 \times 6 = 54 \text{ cm}^2$
- $\text{Perimeter} = 2 \times (9 + 6) = 2 \times 15 = 30 \text{ cm}$

4. $15 \text{ m} \times 10 \text{ m}$

- $\text{Area} = 15 \times 10 = 150 \text{ m}^2$
- $\text{Perimeter} = 2 \times (15 + 10) = 2 \times 25 = 50 \text{ m}$

5. $14 \text{ cm} \times 4 \text{ cm}$

- $\text{Area} = 14 \times 4 = 56 \text{ cm}^2$
- $\text{Perimeter} = 2 \times (14 + 4) = 2 \times 18 = 36 \text{ cm}$

6. $11 \text{ m} \times 9 \text{ m}$

- $\text{Area} = 11 \times 9 = 99 \text{ m}^2$
- $\text{Perimeter} = 2 \times (11 + 9) = 2 \times 20 = 40 \text{ m}$

7. $21 \text{ cm} \times 15 \text{ cm}$

- $\text{Area} = 21 \times 15 = 315 \text{ cm}^2$
- $\text{Perimeter} = 2 \times (21 + 15) = 2 \times 36 = 72 \text{ cm}$

8. $25 \text{ m} \times 13 \text{ m}$

- $\text{Area} = 25 \times 13 = 325 \text{ m}^2$
- $\text{Perimeter} = 2 \times (25 + 13) = 2 \times 38 = 76 \text{ m}$

Worksheet: Find the Missing Length or Width

(Answers)

1. The area of a rectangle is 42 cm^2 and the length is 7 cm.

$$\text{Area} = \text{Length} \times \text{Width}$$

$$42 = 7 \times \text{Width}$$

$$\text{Width} = 6 \text{ cm}$$

2. The perimeter of a rectangle is 30 cm and the width is 5 cm.

$$\text{Perimeter} = 2 \times (\text{Length} + \text{Width})$$

$$30 = 2 \times (\text{Length} + 5)$$

$$15 = \text{Length} + 5$$

$$\text{Length} = 10 \text{ cm}$$

3. A rectangle has an area of 72 m^2 and a width of 8 m.

$$\text{Area} = \text{Length} \times \text{Width}$$

$$72 = \text{Length} \times 8$$

$$\text{Length} = 9 \text{ m}$$

4. The perimeter of a rectangle is 44 cm and the length is 12 cm.

$$\text{Perimeter} = 2 \times (\text{Length} + \text{Width})$$

$$44 = 2 \times (12 + \text{Width})$$

$$22 = 12 + \text{Width}$$

$$\text{Width} = 10 \text{ cm}$$

5. The area of a rectangle is 56 m^2 and one side is 7 m.

$$\text{Area} = \text{Length} \times \text{Width}$$

$$56 = 7 \times \text{other side}$$

$$\text{Other side} = 8 \text{ m}$$

6. A rectangle has a perimeter of 36 cm and a width of 6 cm.

$$\text{Perimeter} = 2 \times (\text{Length} + \text{Width})$$

$$36 = 2 \times (\text{Length} + 6)$$

$$18 = \text{Length} + 6$$

$$\text{Length} = 12 \text{ cm}$$

7. The area of a rectangle is 90 cm^2 and the length is 10 cm.

$$\text{Area} = \text{Length} \times \text{Width}$$

$$90 = 10 \times \text{Width}$$

$$\text{Width} = 9 \text{ cm}$$

8. A rectangle has a perimeter of 50 m and the length is 15 m.

$$\text{Perimeter} = 2 \times (\text{Length} + \text{Width})$$

$$50 = 2 \times (15 + \text{Width})$$

$$25 = 15 + \text{Width}$$

$$\text{Width} = 10 \text{ m}$$

Worksheet: Word Problems

(Answers)

1.

- Area = $12 \text{ m} \times 5 \text{ m} = 60 \text{ m}^2$
- Perimeter = $2 \times (12 + 5) = 2 \times 17 = 34 \text{ m}$

2.

- Area = $10 \text{ cm} \times 6 \text{ cm} = 60 \text{ cm}^2$
- Perimeter = $2 \times (10 + 6) = 2 \times 16 = 32 \text{ cm}$

3.

- Area = $15 \text{ ft} \times 8 \text{ ft} = 120 \text{ ft}^2$
- Perimeter = $2 \times (15 + 8) = 2 \times 23 = 46 \text{ ft}$

4.

- Area = $25 \text{ m} \times 10 \text{ m} = 250 \text{ m}^2$
- Perimeter = $2 \times (25 + 10) = 2 \times 35 = 70 \text{ m}$

5.

- Area = $20 \text{ m} \times 10 \text{ m} = 200 \text{ m}^2$
- Perimeter = $2 \times (20 + 10) = 2 \times 30 = 60 \text{ m}$

6.

- Area = $9 \text{ ft} \times 4 \text{ ft} = 36 \text{ ft}^2$
- Perimeter = $2 \times (9 + 4) = 2 \times 13 = 26 \text{ ft}$

7.

- Area = $7 \text{ ft} \times 3 \text{ ft} = 21 \text{ ft}^2$
- Perimeter = $2 \times (7 + 3) = 2 \times 10 = 20 \text{ ft}$

8.

- Area = $30 \text{ m} \times 20 \text{ m} = 600 \text{ m}^2$
- Perimeter = $2 \times (30 + 20) = 2 \times 50 = 100 \text{ m}$